

(سارا ربیعہ) (امتحان پائینف) (ریاضی محوس 2) (استاد غلامی)
(کلاس: دوشنبہ صا 12 اس 15)

$A(1, -1, 1) \quad B(0, 1, 0) \quad C(1, 0, -1)$

$\vec{AB} = B - A = (0 - 1, 1 - (-1), 0 - 1) = (-1, 2, -1)$

$\vec{AC} = C - A = (1 - 1, 0 - (-1), -1 - 1) = (0, 1, -2)$

$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & -1 \\ 0 & 1 & -2 \end{vmatrix} \rightarrow n = \hat{i}(2(-2) - 0(1)) - \hat{j}((-1)(-2) - 0(0)) + \hat{k}((-1)(1) - 2(0))$
 $n = -4\hat{i} - 2\hat{j} - 1\hat{k} = (-4, -2, -1) \rightarrow n(4, 2, 1)$

$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$
 $4(x - 1) + 2(y - (-1)) + 1(z - 1) = 0 \rightarrow 4(x - 1) + 2(y + 1) + (z - 1) = 0$

$4x - 4 + 2y + 2 + z - 1 = 0 \rightarrow 4x + 2y + z - 3 = 0 \rightarrow 4x + 2y + z = 3$

a) $20x^2 + 15y^2 + 2z^2 = 100$

$\frac{20x^2}{100} + \frac{15y^2}{100} + \frac{2z^2}{100} = \frac{100}{100}$

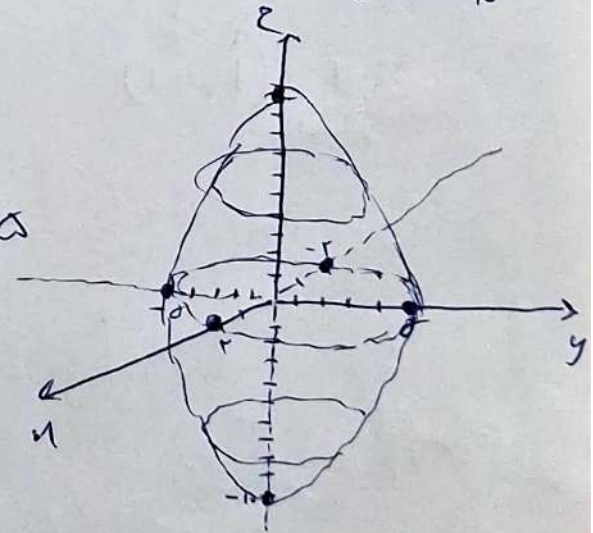
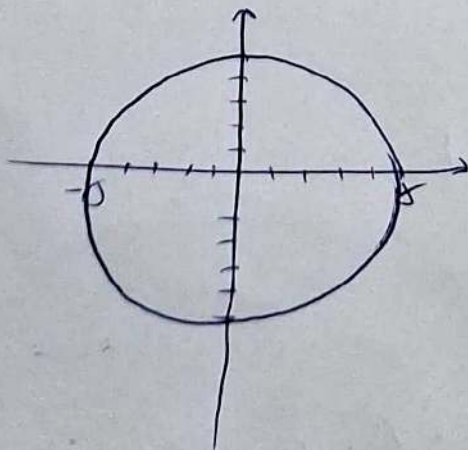
$\Rightarrow \frac{x^2}{5} + \frac{y^2}{20} + \frac{z^2}{50} = 1 \Rightarrow \frac{x^2}{5} + \frac{y^2}{20} + \frac{z^2}{50} = 1$

$\frac{x^2}{5} = 1 \Rightarrow x = \pm \sqrt{5}$

$\frac{y^2}{20} = 1 \Rightarrow y = \pm \sqrt{20}$

$\frac{z^2}{50} = 1 \Rightarrow z = \pm \sqrt{50}$

b) $x^2 + y^2 = 20 \rightarrow r^2 = 20 \rightarrow r = \sqrt{20} = a$



$$K = \frac{\| \vec{R}(t) \times \vec{R}'(t) \|}{\| \vec{R}'(t) \|^2}$$

(3)

$$\frac{d}{dt} (r \sin \theta) = \epsilon \cos \theta$$

$$\frac{d}{dt} (r \cos \theta) = -\epsilon \sin \theta$$

$$\frac{d}{dt} (r) = r'$$

$$\vec{R}'(t) = \epsilon \cos \theta \hat{i} - (\epsilon \sin \theta) \hat{j} + r' \hat{k}$$

$$\| \vec{R}'(t) \| = \sqrt{(\epsilon \cos \theta)^2 + (-\epsilon \sin \theta)^2 + r'^2}$$

$$\| \vec{R}(t) \| = \sqrt{14 \cos^2 \theta + 14 \sin^2 \theta + 9}$$

$$\| \vec{R}(t) \| = \sqrt{14 (\cos^2 \theta + \sin^2 \theta) + 9} = \sqrt{14(1) + 9} = \sqrt{23} = \sigma$$

$$\rightarrow \| \vec{R}(t) \| = \sigma$$

differentiate $\rightarrow \frac{d}{dt} (\epsilon \cos \theta) = -\epsilon \sin \theta$

$$\frac{d}{dt} (-\epsilon \sin \theta) = -\epsilon \cos \theta$$

$$\frac{d}{dt} (r') = 0$$

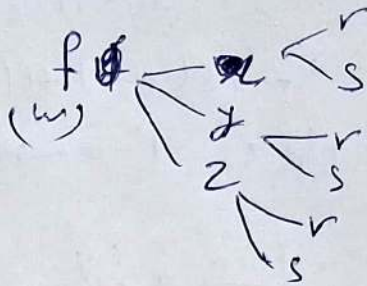
$$\vec{R}''(t) = (-\epsilon \sin \theta) \hat{i} - (\epsilon \cos \theta) \hat{j} + 0 \hat{k}$$

$$\vec{R}(t) \times \vec{R}''(t) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \epsilon \cos \theta & -\epsilon \sin \theta & r' \\ -\epsilon \sin \theta & -\epsilon \cos \theta & 0 \end{vmatrix} \xrightarrow{\text{Simplify}} \vec{R}(t) \times \vec{R}''(t) = (\epsilon \cos \theta) \hat{i} - (\epsilon \sin \theta) \hat{j} - r' \hat{k}$$

$-r' \hat{k}$

$\xrightarrow{\text{Simplify}} \left[K = \frac{\epsilon}{r\omega} \right]$

$$\begin{cases} w = r^r + y^r + z^r \\ n = r - s \\ y = \cos(r+s) \\ z = \sin(r+s) \end{cases}$$



(f)

$$\frac{\partial w}{\partial r} = \frac{\partial f}{\partial r}$$

$\begin{matrix} \vec{r} & \vec{s} \\ \downarrow & \downarrow \\ (1, -1) \end{matrix}$

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial n} \frac{\partial n}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r}$$

$\begin{matrix} \text{نات } n & \text{نات } r & \text{نات } y & \text{نات } z \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \frac{\partial n}{\partial r} & \frac{\partial y}{\partial r} & \frac{\partial z}{\partial r} \end{matrix}$

$$f_n = r^n \quad f_s = -r$$

$$f_y = r^r \quad f_s = -\sin(r+s)$$

$$f_z = r^r \quad f_s = \cos(r+s)$$

$$r^n(-r) + r^r(-\sin(r+s)) + r^r(\cos(r+s)) =$$

$$(1)(1) + 1(0) + (0)(1) \rightarrow \frac{\partial w}{\partial r} = 1 + 0 + 0 = 1$$

$$f(x, y, z) = x^2 - xy^2 - z^2$$

$$P(1, 1, 0)$$

$$\vec{r} = x\vec{i} - y\vec{j} + z\vec{k}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2} = r \Rightarrow \vec{u} = \frac{\vec{r}}{|\vec{r}|} = \frac{x}{r}\vec{i} - \frac{y}{r}\vec{j} + \frac{z}{r}\vec{k}$$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (2x - y^2, -2xy, -2z)$$

$$\nabla f(1, 1, 0) = 2\vec{i} - 2\vec{j} - 2\vec{k} \quad \frac{\vec{r}}{r} \times \vec{r} = \frac{\vec{r}}{r}$$

$$D_u f(1, 1, 0) = \nabla f(1, 1, 0) \cdot \vec{u} = \frac{2}{r} + \frac{2}{r} - \frac{2}{r} = \frac{2}{r}$$

$$f(x, y) = x^2 + y^2 - \epsilon xy$$

$$f_x = \frac{\partial f}{\partial x} = 2x - \epsilon y$$

$$f_y = \frac{\partial f}{\partial y} = 2y - \epsilon x$$

$$\begin{cases} 2x - \epsilon y = 0 \rightarrow y = \frac{2x}{\epsilon} \\ 2y - \epsilon x = 0 \rightarrow x = \frac{2y}{\epsilon} \end{cases}$$

$$x = (x')^2 \Rightarrow x = x' \rightarrow x' - x = 0 \rightarrow x'(x' - 1) = 0$$

$$\begin{cases} f_{xx} = 2 \\ f_{yy} = 2 \\ f_{xy} = -\epsilon \end{cases}$$

$$D(x, y) = f_{xx}f_{yy} - (f_{xy})^2 =$$

$$(2)(2) - (-\epsilon)^2 = 4 - \epsilon^2$$

$$D(0, 0) = 4 - \epsilon^2 = 4$$

$$D(1, 1) = 4 - \epsilon^2 = 4 - \epsilon^2$$

$$D(-1, -1) = 4 - \epsilon^2 = 4 - \epsilon^2$$

$$\begin{aligned} x=0, y=0 & \rightarrow (0, 0) \\ x=1, y=1 & \rightarrow (1, 1) \\ x=-1, y=-1 & \rightarrow (-1, -1) \end{aligned}$$

$$\int_0^1 \int_{\sqrt[r]{y}}^r \frac{1}{x^{\epsilon+1}} dx dy$$

مستطيل

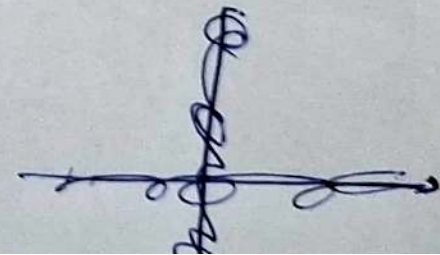
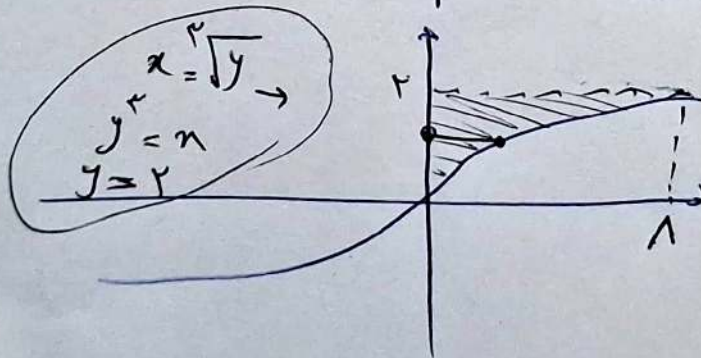
$$\int \sqrt[r]{y} \leq x \leq r \rightarrow x = \sqrt[r]{y} \rightarrow y = x^r$$

$0 \leq y \leq 1 \xrightarrow{y=0} x=0$
 $y=r \rightarrow x=r$

$$\int_0^r \int_0^{x^r} \frac{1}{x^{\epsilon+1}} dy dx \rightarrow \int_0^r \left(\frac{y}{x^{\epsilon+1}} \right) \Big|_0^{x^r} dx$$

$$\rightarrow \int_0^r \left(\frac{x^r}{x^{\epsilon+1}} - 0 \right) dx = \int_0^r \frac{x^r}{x^{\epsilon+1}} dx$$

$$= \frac{1}{\epsilon} \ln(x^{\epsilon} + 1) \Big|_0^r = \frac{1}{\epsilon} (\ln 1^{\epsilon} - \ln 1) = \frac{1}{\epsilon} \ln 1^{\epsilon}$$



$$xy = v / xy = e / \frac{y^r}{x} = \theta, \frac{y^r}{x} = 1$$

(1)

$$u = xy \rightarrow e \leq u \leq v$$

$$v = \frac{y^r}{x} \rightarrow 1 \leq v \leq \theta$$

$$J^{-1} = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} y & x \\ \frac{y^r}{x^r} & \frac{ry}{x} \end{vmatrix}$$

$$\xrightarrow{\text{دائره مبركبات}} |J^{-1}| = rv \rightarrow |J| = \frac{1}{|J^{-1}|} = \left(\frac{1}{rv} \right)$$

$$A = \int_1^{\theta} \int_1^v |J| du dv \rightarrow A = \int_1^{\theta} \int_1^v \frac{1}{rv} du dv \rightarrow$$

$$\xrightarrow{\text{انكشاف بنيت}} \int_1^v du = v - e = r \rightarrow$$

$$A = \int_1^{\theta} r \times \frac{1}{rv} dv = \int_1^{\theta} \frac{1}{v} dv \rightarrow A = \ln(v) \Big|_1^{\theta} = \ln(\theta) - \ln(1)$$

$$\Rightarrow A = \ln(\theta)$$

~~Problem 1~~

$$V = \int_0^{2\pi} \int_1^2 \int_0^{\sqrt{\epsilon - z^2}} r \, dr \, dz \, d\theta$$

(9)

(r, θ, z)

$$r^2 + z^2 = \epsilon \rightarrow r^2 = \epsilon - z^2$$

$$1 \leq z \leq 2$$

$$r \leq \sqrt{\epsilon - z^2}$$

$$0 \leq \theta \leq 2\pi$$

$$dV = r \, dr \, dz \, d\theta$$

$$\int_0^{\sqrt{\epsilon - z^2}} r \, dr = \left. \frac{r^2}{2} \right|_0^{\sqrt{\epsilon - z^2}} = \frac{\epsilon - z^2}{2}$$

$$\int_1^2 \frac{\epsilon - z^2}{2} \, dz = \frac{1}{2} \left(\epsilon z - \frac{z^3}{3} \right) \Big|_1^2$$

$$\frac{1}{2} \left(\frac{14}{3} - \frac{11}{3} \right) = \frac{1}{2} \left(\frac{\phi}{3} \right) = \frac{\phi}{6}$$

$$z=2 \rightarrow \epsilon(r) - \frac{r^2}{2} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$z=1 \rightarrow \epsilon(1) - \frac{1^2}{2} = \epsilon - \frac{1}{2} = \frac{11}{2}$$

$$V = \int_0^{2\pi} \frac{\phi}{6} \, d\theta = \frac{\phi}{6} \cdot 2\pi = \frac{\phi\pi}{3}$$

$$\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA$$

(10)

~~Problem 2~~ $\rightarrow P = y^r$
 $Q = x^r$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x} (x^r) = r x^{r-1}$$

$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y} (y^r) = r y^{r-1}$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = r x^{r-1} - r y^{r-1}$$

$x=0$
 $y=0$
 $x+y=1 \rightarrow y=1-x$

$$\iint_D (r x^{r-1} - r y^{r-1}) \, dA = \int_0^1 \int_0^{1-x} (r x^{r-1} - r y^{r-1}) \, dy \, dx \rightarrow \int_0^1 (r x^{r-1} (1-x) - \frac{r}{2} (1-x)^2) \, dx$$

$$= \left[r x^r (1-x) - \frac{r}{2} (1-x)^2 \right]_0^1 = r(1-1) - \frac{r}{2}(0) = 0$$

$$\int_0^1 (-3x^2 + 2x^2 + 2x - 1) dx = \left(-\frac{3}{3}x^3 + \frac{2}{2}x^2 + 2x - x \right) \Big|_0^1 \quad \leftarrow \textcircled{1 \rightarrow b)}$$

$$= -\frac{3}{3} (1)^3 + \frac{2}{2} (1)^2 + (1)^2 - 1 = -\frac{3}{3} + \frac{2}{2} + 1 - 1$$

$$= -\frac{3}{3} + \frac{2}{2} = \frac{-9 + 1}{12} = \frac{-8}{12} = -\frac{2}{3}$$